



2026

PHYSICS

Predicted Paper 4 of 4 – Radioactivity • Thermal Physics • Transformers

Time allowed: 45 minutes Total marks: 50

Candidate name: _____

Class / Grade: _____ Date: _____

Instructions to candidates

1. Answer **all** questions in Sections A, B and C.
2. Section A: place a tick (✓) in the box next to your chosen answer.
3. Section B: write answers in the spaces provided. Show all working.
4. Section C: answer both structured questions. Show full working for calculations.
5. Write clearly in blue or black ink. Pencil may be used for diagrams.
6. Marks for each question are shown in **[brackets]**.
7. Do not write in the marks column (examiner use only).

Formula Sheet

Thermal energy: $Q = mc\Delta T$ | Specific heat capacity of water: $c = 4200 \text{ J/(kg}\cdot\text{K)}$

Gas laws (fixed mass): $pV/T = \text{constant}$ | Boyle: $p_1V_1 = p_2V_2$ (constant T) |

Charles: $V_1/T_1 = V_2/T_2$ (constant p)

Transformer: $V_p/V_s = N_p/N_s$ | $V_pI_p = V_sI_s$ (100% efficiency)

Half-life: after n half-lives, activity = initial $\times (1/2)^n$

Ohm's law: $V = IR$ | Power: $P = IV = I^2R$

$g = 10 \text{ N/kg}$ | $0^\circ\text{C} = 273 \text{ K}$

Section	Marks available	Marks awarded
A (MCQ)	15	
B (Short structured)	20	

C (Long structured)	15	
Total	50	

SECTION A – Multiple Choice [15 marks]

Answer ALL 15 questions. Tick

ONE box per question.

1. Alpha radiation is stopped by: [1]
- ☐ A A few centimetres of air
- ☐ B A sheet of paper or skin
- ☐ C A few millimetres of aluminium
- ☐ D Several centimetres of lead
2. A radioactive sample has a half-life of 20 minutes. Starting with 800 counts/s, after 60 minutes the count rate will be: [1]
- ☐ A 400 counts/s
- ☐ B 200 counts/s
- ☐ C 100 counts/s
- ☐ D 50 counts/s
3. Beta particles are: [1]
- ☐ A Helium nuclei (2 protons + 2 neutrons)
- ☐ B High-energy electromagnetic radiation
- ☐ C Fast-moving electrons emitted from the nucleus
- ☐ D Positive ions
4. Which type of radiation has the greatest penetrating power? [1]
- ☐ A Alpha
- ☐ B Beta
- ☐ C Gamma
- ☐ D All three are equal
5. A use of gamma radiation in medicine is: [1]
- ☐ A Detecting leaks in underground pipes
- ☐ B Sterilising medical equipment
- ☐ C Measuring the thickness of paper
- ☐ D Powering smoke detectors
6. Heat energy required to raise 2 kg of water by 5 °C ($c = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$) is: [1]
- ☐ A 4200 J

- ☐ B 8400 J
- ☐ C 42 000 J
- ☐ D 21 000 J

7. As you heat a gas at constant volume, the pressure:

[1]

- ☐ A Decreases
- ☐ B Stays the same
- ☐ C Increases
- ☐ D First increases then decreases

8. A gas occupies 4 L at 200 kPa. At constant temperature, if the pressure doubles to 400 kPa, the new volume is:

[1]

- ☐ A 8 L
- ☐ B 4 L
- ☐ C 2 L
- ☐ D 1 L

9. The Kelvin temperature of 27 °C is:

[1]

- ☐ A 27 K
- ☐ B 246 K
- ☐ C 300 K
- ☐ D 373 K

10. A step-down transformer is used to:

[1]

- ☐ A Increase voltage and decrease current
- ☐ B Decrease voltage and increase current
- ☐ C Increase both voltage and current
- ☐ D Convert d.c. to a.c.

11. Background radiation comes from all of the following EXCEPT:

[1]

- ☐ A Cosmic rays
- ☐ B Radon gas
- ☐ C Rocks and soil
- ☐ D Sunlight

12. The specific heat capacity of a material is the energy needed to raise:

[1]

- ☐ A 1 kg of the material by 1 °C (or 1 K)
- ☐ B 1 g of the material by 100 °C

- ☐ C Any mass of the material by 1 °C
- ☐ D 1 kg of any material by 1 °C

13. Why are power lines operated at very high voltage?

[1]

- ☐ A To increase the resistance of the cables
- ☐ B To reduce the current and so reduce energy lost as heat in the cables
- ☐ C To increase the power delivered
- ☐ D To make the cables lighter

14. During alpha decay the nucleus loses:

[1]

- ☐ A 1 proton and 1 neutron
- ☐ B 2 protons and 2 neutrons
- ☐ C 1 electron
- ☐ D No nucleons — only energy is released

15. Which statement best explains why gas pressure increases when temperature rises (constant volume)?

[1]

- ☐ A Gas molecules become larger
- ☐ B Gas molecules collide with the walls more frequently and with greater force
- ☐ C Gas molecules slow down
- ☐ D The number of molecules increases

SECTION B – Short Structured Questions [20 marks] Answer ALL 4 questions.

16. Types of radioactive emission

The table below shows three types of nuclear radiation. Complete it.

Radiation	Nature	Charge	Stopped by
α	Helium nucleus	+2	
β		-1	Few mm of aluminium
γ	Electromagnetic wave		Several cm of lead

[3]

- (b) A radioactive isotope has a half-life of 30 years. A sample currently has an activity of 3200 Bq. Calculate the activity after 90 years.

[2]

Activity = _____ Bq

17. Half-life and decay curve

The graph below shows the decay of a radioactive sample.

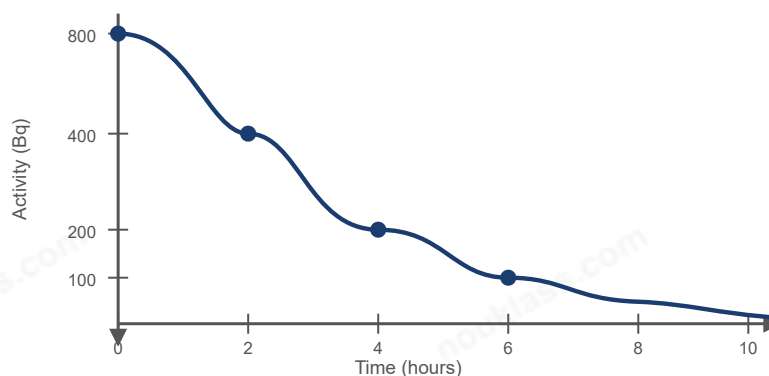


Fig. 17.1 – Activity vs time for a radioactive sample

- (a) Use the graph to determine the half-life of this sample. Show your method on the graph (draw construction lines).

[2]

Half-life = _____ hours

- (b) State the initial activity of the sample. [1]

Initial activity = _____ Bq

- (c) Suggest one precaution a scientist handling this sample should take and give a reason. [2]

18. Specific heat capacity

A kettle heats 0.5 kg of water from 20 °C to 100 °C. ($c_{\text{water}} = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$)

- (a) Calculate the energy needed to heat the water. [2]

$Q =$ _____ J

- (b) The kettle has a power rating of 2100 W. Calculate the minimum time to heat the water. [2]

Time = _____ s

- (c) In practice the kettle takes longer than your calculated time. Give one reason. [1]

19. Gas laws

- (a) A gas has volume 3.0 L at a pressure of 100 kPa. The temperature is kept constant and the pressure is increased to 150 kPa. Calculate the new volume. [2]

New volume = _____ L

- (b) Explain, using the kinetic model, why the pressure of a gas in a sealed container increases when the gas is heated. [2]

- (c) Convert 127 °C to Kelvin. [1]

= _____ K

SECTION C – Long Structured Questions [15 marks] Answer BOTH questions.

20. Transformers and the National Grid

A power station generates electricity at 25 000 V. A step-up transformer raises this to 400 000 V for long-distance transmission. At the consumer end, a step-down transformer reduces it to 240 V for homes.

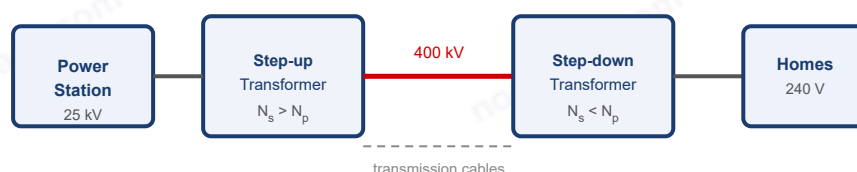


Fig. 20.1 – Simplified National Grid

- (a) The step-up transformer has 500 turns on the primary. Calculate the number of turns on the secondary coil. [2]

$N_s =$ _____ turns

- (b) The step-up transformer delivers a current of 50 A at 400 000 V. The power station generates at 25 000 V. Assuming 100% efficiency, calculate the current at the power station output. [2]

$I_p =$ _____ A

- (c) The step-down transformer reduces 400 000 V to 240 V. If it has 20 000 turns on the primary, calculate the number of turns on the secondary. [2]

$N_s = \underline{\hspace{2cm}}$ turns

- (d) Explain why high-voltage transmission reduces energy losses in the cables. Use the formula $P = I^2R$ in your answer. [2]

21. Thermal physics — heating water and gas laws

- (a) A 3 kW immersion heater heats 4 kg of water. The water starts at 15 °C.
(i) Calculate the energy needed to heat the water to 75 °C. ($c = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$) [2]

$Q = \underline{\hspace{2cm}}$ J

- (ii) Calculate the time the heater takes (assume 100% efficiency). [2]

t = _____ s (= _____ min)

- (b) A sealed container holds gas at 200 kPa and 300 K. The temperature rises to 450 K at constant volume. Calculate the new pressure. [3]



New pressure = _____ kPa

Mark Scheme – Predicted Paper 4 | Physics NCE 2026

For **teacher and self-assessment use only**. Equivalent correct answers should receive full credit. Award method marks where an arithmetic slip causes the final answer to be wrong but the method is correct.

Section A – MCQ Answers

Q	Answer	Key reasoning
1	B	Alpha is stopped by a sheet of paper or a few cm of air (very short range)
2	C	60 min = 3 half-lives; $800 \rightarrow 400 \rightarrow 200 \rightarrow 100$ counts/s
3	C	Beta: fast electrons emitted from the nucleus when a neutron decays to a proton
4	C	Gamma has greatest penetrating power; requires thick lead to reduce
5	B	Gamma is used to sterilise medical equipment (destroys bacteria DNA)
6	C	$Q = mc\Delta T = 2 \times 4200 \times 5 = 42\,000$ J
7	C	Higher T $\rightarrow$ faster molecules $\rightarrow$ more frequent, harder collisions $\rightarrow$ higher pressure
8	C	Boyle's law: $p_1V_1 = p_2V_2$; $200 \times 4 = 400 \times V_2$; $V_2 = 2$ L
9	C	$T(K) = 27 + 273 = 300$ K
10	B	Step-down: more turns on primary $\rightarrow$ lower secondary voltage, higher current
11	D	Sunlight is not a source of background radiation (it is electromagnetic, not ionising in ordinary amounts)
12	A	SHC = energy to raise 1 kg by 1 K (definition)
13	B	$P = IV$; same P at higher V means lower I; power loss = I^2R , so lower I greatly reduces losses
14	B	Alpha decay: nucleus loses 2 protons + 2 neutrons (helium-4 nucleus)
15	B	Kinetic model: hotter gas $\rightarrow$ faster molecules $\rightarrow$ more frequent and harder wall collisions $\rightarrow$ higher pressure

Section B

Q	Part	Mark points	Marks
16	(table)	α stopped by: paper / few cm of air [1]; β nature: fast electrons [1]; γ charge: 0 (neutral) [1]	3

	(b)	90 years = 3 half-lives [1]; $3200 \rightarrow 1600 \rightarrow 800 \rightarrow \mathbf{400 \text{ Bq}}$ [1]	2
17	(a)	Read off: activity halves from 800 to 400 Bq between $t = 0$ and $t = 2 \text{ h}$ [1]; half-life = 2 hours [1]	2
	(b)	Initial activity = 800 Bq	1
	(c)	Precaution (e.g. use tongs / wear gloves) [1]; reason (e.g. to avoid skin contact with ionising radiation which can damage cells) [1]	2
18	(a)	$Q = mc\Delta T = 0.5 \times 4200 \times (100 - 20) = 0.5 \times 4200 \times 80 = \mathbf{168\,000 \text{ J}}$	2
	(b)	$t = Q/P = 168\,000/2100 = \mathbf{80 \text{ s}}$	2
	(c)	Energy is lost heating the kettle body itself / energy lost to surroundings / heating element not 100% efficient [1 — any one]	1
19	(a)	Boyle's law: $p_1V_1 = p_2V_2$; $100 \times 3.0 = 150 \times V_2$; $V_2 = \mathbf{2.0 \text{ L}}$	2
	(b)	Heating gives molecules more kinetic energy [1]; they move faster and collide with the container walls more frequently and with greater force, so pressure rises [1]	2
	(c)	$127 + 273 = \mathbf{400 \text{ K}}$	1

Section C

Q	Part	Mark points	Marks
20	(a)	$N_s/N_p = V_s/V_p$; $N_s = 500 \times 400\,000/25\,000 = \mathbf{8000 \text{ turns}}$	2
	(b)	$V_p I_p = V_s I_s$; $I_p = 400\,000 \times 50 / 25\,000 = \mathbf{800 \text{ A}}$	2
	(c)	$N_s = 20\,000 \times 240/400\,000 = \mathbf{12 \text{ turns}}$	2
	(d)	High voltage $\rightarrow$ low current (same power $P = IV$) [1]; power loss = I^2R , so lower current greatly reduces heating losses in cables [1]	2
21	(a) (i)	$Q = mc\Delta T = 4 \times 4200 \times 60 = \mathbf{1\,008\,000 \text{ J}}$ (accept 1.008 MJ)	2
	(a) (ii)	$t = Q/P = 1\,008\,000/3000 = \mathbf{336 \text{ s}}$ (= 5 min 36 s)	2
	(b)	Constant volume: $p/T = \text{constant}$; $p_2 = p_1 \times T_2/T_1$ [1]; = $200 \times 450/300$ [1] = 300 kPa [1]	3